

FagusED · Biology

01 · Characteristics and classification of living organisms

Define: species.

BIO-0001

PREGUNTA

FagusED · Biology

01 · Characteristics and classification of living organisms

Define: binomial system.

BIO-0002

PREGUNTA

FagusED · Biology

01 · Characteristics and classification of living organisms

Define: dichotomous key.

BIO-0003

PREGUNTA

FagusED · Biology

01 · Characteristics and classification of living organisms

Define: vertebrate.

BIO-0004

PREGUNTA

FagusED · Biology

01 · Characteristics and classification of living organisms

Define: virus.

BIO-0005

PREGUNTA

FagusED · Biology

01 · Characteristics and classification of living organisms

Explain the key idea of characteristics of life.

BIO-0006

PREGUNTA

FagusED · Biology

01 · Characteristics and classification of living organisms

Explain the key idea of classification.

BIO-0007

PREGUNTA

FagusED · Biology

01 · Characteristics and classification of living organisms

Apply the concept: a newly found organism has chitin cell walls and absorbs digested nutrients. What would you conclude?

BIO-0008

PREGUNTA

FagusED · Biology

02 · Organisation of the organism

Define: tissue.

BIO-0009

PREGUNTA

FagusED · Biology

02 · Organisation of the organism

Define: organ.

BIO-0010

PREGUNTA

FagusED · Biology

02 · Organisation of the organism

Define: organelle.

BIO-0011

PREGUNTA

FagusED · Biology

02 · Organisation of the organism

Define: magnification.

BIO-0012

PREGUNTA

FagusED · Biology

02 · Organisation of the organism

Define: resolution.

BIO-0013

PREGUNTA

FagusED · Biology

02 · Organisation of the organism

Explain the key idea of cell structures.

BIO-0014

PREGUNTA

FagusED · Biology

02 · Organisation of the organism

Explain the key idea of specialised cells.

BIO-0015

PREGUNTA

FagusED · Biology**01 · Characteristics and classification of living organisms**

a sequence of paired contrasting statements used for identification

BIO-0003

RESPUESTA

FagusED · Biology**01 · Characteristics and classification of living organisms**

a two-part scientific name consisting of genus and species

BIO-0002

RESPUESTA

FagusED · Biology**01 · Characteristics and classification of living organisms**

a group of organisms that can reproduce to produce fertile offspring

BIO-0001

RESPUESTA

FagusED · Biology**01 · Characteristics and classification of living organisms**

Living organisms show movement, respiration, sensitivity, growth, reproduction, excretion and nutrition. These characteristics describe coordinated cellular processes; no single characteristic alone is always sufficient to decide whether an unfamiliar object is living.

BIO-0006

RESPUESTA

FagusED · Biology**01 · Characteristics and classification of living organisms**

genetic material enclosed in a protein coat that replicates only in host cells

BIO-0005

RESPUESTA

FagusED · Biology**01 · Characteristics and classification of living organisms**

an animal with a vertebral column

BIO-0004

RESPUESTA

FagusED · Biology**02 · Organisation of the organism**

a group of cells with similar structures working together for a function

BIO-0009

RESPUESTA

FagusED · Biology**01 · Characteristics and classification of living organisms**

classify it as a fungus. chitin walls and absorptive nutrition are characteristic of fungi

BIO-0008

RESPUESTA

FagusED · Biology**01 · Characteristics and classification of living organisms**

Classification places organisms into groups using shared features. Modern systems also use DNA base-sequence evidence to infer evolutionary relationships. Binomial naming gives each species a two-part Latin name: genus followed by species.

BIO-0007

RESPUESTA

FagusED · Biology**02 · Organisation of the organism**

image size divided by actual size

BIO-0012

RESPUESTA

FagusED · Biology**02 · Organisation of the organism**

a specialised structure within a cell

BIO-0011

RESPUESTA

FagusED · Biology**02 · Organisation of the organism**

a structure made of different tissues working together

BIO-0010

RESPUESTA

FagusED · Biology**02 · Organisation of the organism**

Structure is adapted to function: root hair cells have a long extension for absorption; red blood cells are biconcave and lack a nucleus; sperm cells have a flagellum and many mitochondria; palisade cells contain many chloroplasts; ciliated cells move mucus.

BIO-0015

RESPUESTA

FagusED · Biology**02 · Organisation of the organism**

The cell membrane controls movement of substances; cytoplasm is the site of many reactions; the nucleus contains genetic material; mitochondria carry out aerobic respiration; ribosomes synthesise proteins. Plant cells additionally have a cellulose wall, permanent vacuole and chloroplasts in photosynthetic tissues.

BIO-0014

RESPUESTA

FagusED · Biology**02 · Organisation of the organism**

the ability to distinguish two close points as separate

BIO-0013

RESPUESTA

FagusED · Biology
02 · Organisation of the organism

Apply the concept: a micrograph shows a cell image 40 mm long at magnification x800. What would you conclude?

BIO-0016

PREGUNTA

FagusED · Biology
03 · Movement into and out of cells

Define: diffusion.

BIO-0017

PREGUNTA

FagusED · Biology
03 · Movement into and out of cells

Define: osmosis.

BIO-0018

PREGUNTA

FagusED · Biology
03 · Movement into and out of cells

Define: active transport.

BIO-0019

PREGUNTA

FagusED · Biology
03 · Movement into and out of cells

Define: turgid.

BIO-0020

PREGUNTA

FagusED · Biology
03 · Movement into and out of cells

Define: plasmolysis.

BIO-0021

PREGUNTA

FagusED · Biology
03 · Movement into and out of cells

Explain the key idea of diffusion.

BIO-0022

PREGUNTA

FagusED · Biology
03 · Movement into and out of cells

Explain the key idea of osmosis.

BIO-0023

PREGUNTA

FagusED · Biology
03 · Movement into and out of cells

Apply the concept: root cells contain more nitrate ions than surrounding soil. What would you conclude?

BIO-0024

PREGUNTA

FagusED · Biology
04 · Biological molecules

Define: reducing sugar.

BIO-0025

PREGUNTA

FagusED · Biology
04 · Biological molecules

Define: protein.

BIO-0026

PREGUNTA

FagusED · Biology
04 · Biological molecules

Define: lipid.

BIO-0027

PREGUNTA

FagusED · Biology
04 · Biological molecules

Define: gene.

BIO-0028

PREGUNTA

FagusED · Biology
04 · Biological molecules

Define: nucleotide.

BIO-0029

PREGUNTA

FagusED · Biology
04 · Biological molecules

Explain the key idea of carbohydrates.

BIO-0030

PREGUNTA

FagusED · Biology**03 · Movement into and out of cells**

net movement of water from higher to lower water potential through a partially permeable membrane

BIO-0018

RESPUESTA

FagusED · Biology**03 · Movement into and out of cells**

net movement from higher to lower concentration due to random motion

BIO-0017

RESPUESTA

FagusED · Biology**02 · Organisation of the organism**

actual length is 0.050 mm or 50 micrometres. actual size equals image size divided by magnification after unit conversion

BIO-0016

RESPUESTA

FagusED · Biology**03 · Movement into and out of cells**

the cell membrane pulling away from the wall after water loss

BIO-0021

RESPUESTA

FagusED · Biology**03 · Movement into and out of cells**

a plant cell firm from water uptake and pressure against its wall

BIO-0020

RESPUESTA

FagusED · Biology**03 · Movement into and out of cells**

movement against a concentration gradient using energy

BIO-0019

RESPUESTA

FagusED · Biology**03 · Movement into and out of cells**

nitrate uptake occurs by active transport. ions move against the gradient using respiratory energy and membrane carriers

BIO-0024

RESPUESTA

FagusED · Biology**03 · Movement into and out of cells**

Osmosis is the net movement of water molecules from higher water potential to lower water potential through a partially permeable membrane. Dilute solution has higher water potential than concentrated solution.

BIO-0023

RESPUESTA

FagusED · Biology**03 · Movement into and out of cells**

Diffusion is the net movement of particles from higher to lower concentration due to random motion. A steeper concentration gradient, higher temperature and larger surface area increase rate; a shorter distance also increases exchange.

BIO-0022

RESPUESTA

FagusED · Biology**04 · Biological molecules**

a molecule formed from fatty acids and glycerol

BIO-0027

RESPUESTA

FagusED · Biology**04 · Biological molecules**

a polymer of amino acids folded into a specific shape

BIO-0026

RESPUESTA

FagusED · Biology**04 · Biological molecules**

a sugar that reduces Benedict's reagent when heated

BIO-0025

RESPUESTA

FagusED · Biology**04 · Biological molecules**

Glucose is a simple reducing sugar used in respiration. Starch and glycogen are large insoluble storage carbohydrates made from glucose; cellulose forms plant cell walls. Insolubility prevents storage molecules affecting water potential and allows compact storage.

BIO-0030

RESPUESTA

FagusED · Biology**04 · Biological molecules**

a repeating unit of DNA containing a base, sugar and phosphate

BIO-0029

RESPUESTA

FagusED · Biology**04 · Biological molecules**

a length of DNA coding for a protein

BIO-0028

RESPUESTA

FagusED · Biology
04 · Biological molecules

Explain the key idea of proteins and lipids.

BIO-0031

PREGUNTA

FagusED · Biology
04 · Biological molecules

Apply the concept: an unknown food gives lilac Biuret and milky emulsion results. What would you conclude?

BIO-0032

PREGUNTA

FagusED · Biology
05 · Enzymes

Define: enzyme.

BIO-0033

PREGUNTA

FagusED · Biology
05 · Enzymes

Define: active site.

BIO-0034

PREGUNTA

FagusED · Biology
05 · Enzymes

Define: substrate.

BIO-0035

PREGUNTA

FagusED · Biology
05 · Enzymes

Define: denaturation.

BIO-0036

PREGUNTA

FagusED · Biology
05 · Enzymes

Define: optimum.

BIO-0037

PREGUNTA

FagusED · Biology
05 · Enzymes

Explain the key idea of enzyme action.

BIO-0038

PREGUNTA

FagusED · Biology
05 · Enzymes

Explain the key idea of temperature.

BIO-0039

PREGUNTA

FagusED · Biology
05 · Enzymes

Apply the concept: a patient has a fever well above an enzyme optimum. What would you conclude?

BIO-0040

PREGUNTA

FagusED · Biology
06 · Plant nutrition

Define: photosynthesis.

BIO-0041

PREGUNTA

FagusED · Biology
06 · Plant nutrition

Define: limiting factor.

BIO-0042

PREGUNTA

FagusED · Biology
06 · Plant nutrition

Define: stoma.

BIO-0043

PREGUNTA

FagusED · Biology
06 · Plant nutrition

Define: chlorophyll.

BIO-0044

PREGUNTA

FagusED · Biology
06 · Plant nutrition

Define: translocation.

BIO-0045

PREGUNTA

FagusED · Biology
05 · Enzymes

a protein that acts as a biological catalyst

BIO-0033

RESPUESTA

FagusED · Biology
04 · Biological molecules

protein and lipid are present. Biuret identifies protein and the ethanol emulsion identifies lipid

BIO-0032

RESPUESTA

FagusED · Biology
04 · Biological molecules

Proteins are chains of amino acids folded into specific shapes and function as enzymes, antibodies and structural materials. Lipids contain fatty acids and glycerol, store concentrated energy, insulate and form cell membranes.

BIO-0031

RESPUESTA

FagusED · Biology
05 · Enzymes

a permanent change in protein shape that alters the active site

BIO-0036

RESPUESTA

FagusED · Biology
05 · Enzymes

the reactant molecule on which an enzyme acts

BIO-0035

RESPUESTA

FagusED · Biology
05 · Enzymes

the region of an enzyme where the substrate binds

BIO-0034

RESPUESTA

FagusED · Biology
05 · Enzymes

Raising temperature initially increases kinetic energy and successful collisions. Above the optimum, bonds maintaining enzyme shape break, the active site changes and the enzyme denatures; rate falls rapidly. Low temperature slows reactions but does not normally denature.

BIO-0039

RESPUESTA

FagusED · Biology
05 · Enzymes

Enzymes are proteins that catalyse metabolic reactions. A substrate with a complementary shape binds at the active site to form an enzyme-substrate complex; products leave and the enzyme can be reused. Enzymes lower activation energy.

BIO-0038

RESPUESTA

FagusED · Biology
05 · Enzymes

the condition at which enzyme rate is greatest

BIO-0037

RESPUESTA

FagusED · Biology
06 · Plant nutrition

a factor in shortest supply that restricts reaction rate

BIO-0042

RESPUESTA

FagusED · Biology
06 · Plant nutrition

the process that makes carbohydrates from carbon dioxide and water using light energy

BIO-0041

RESPUESTA

FagusED · Biology
05 · Enzymes

reaction rate may decrease. high temperature disrupts bonds, changes the active site and reduces enzyme-substrate complexes

BIO-0040

RESPUESTA

FagusED · Biology
06 · Plant nutrition

movement of sucrose and amino acids in phloem

BIO-0045

RESPUESTA

FagusED · Biology
06 · Plant nutrition

a green pigment that absorbs light energy

BIO-0044

RESPUESTA

FagusED · Biology
06 · Plant nutrition

a pore in the leaf epidermis controlled by guard cells

BIO-0043

RESPUESTA

FagusED · Biology
06 · Plant nutrition

Explain the key idea of photosynthesis.

BIO-0046

PREGUNTA

FagusED · Biology
06 · Plant nutrition

Explain the key idea of leaf adaptations.

BIO-0047

PREGUNTA

FagusED · Biology
06 · Plant nutrition

Apply the concept: a plant has yellow leaves but adequate light and water. What would you conclude?

BIO-0048

PREGUNTA

FagusED · Biology
07 · Human nutrition

Define: ingestion.

BIO-0049

PREGUNTA

FagusED · Biology
07 · Human nutrition

Define: digestion.

BIO-0050

PREGUNTA

FagusED · Biology
07 · Human nutrition

Define: absorption.

BIO-0051

PREGUNTA

FagusED · Biology
07 · Human nutrition

Define: assimilation.

BIO-0052

PREGUNTA

FagusED · Biology
07 · Human nutrition

Define: egestion.

BIO-0053

PREGUNTA

FagusED · Biology
07 · Human nutrition

Explain the key idea of balanced diet.

BIO-0054

PREGUNTA

FagusED · Biology
07 · Human nutrition

Explain the key idea of digestion.

BIO-0055

PREGUNTA

FagusED · Biology
07 · Human nutrition

Apply the concept: a person has flattened intestinal villi. What would you conclude?

BIO-0056

PREGUNTA

FagusED · Biology
08 · Transport in plants

Define: transpiration.

BIO-0057

PREGUNTA

FagusED · Biology
08 · Transport in plants

Define: translocation.

BIO-0058

PREGUNTA

FagusED · Biology
08 · Transport in plants

Define: xylem.

BIO-0059

PREGUNTA

FagusED · Biology
08 · Transport in plants

Define: phloem.

BIO-0060

PREGUNTA

FagusED · Biology
06 · Plant nutrition

magnesium deficiency is likely. magnesium is required to make chlorophyll, so deficiency reduces green pigment and photosynthesis

BIO-0048

RESPUESTA

FagusED · Biology
06 · Plant nutrition

A broad thin lamina gives large area and short diffusion distance. Palisade cells near the upper surface contain many chloroplasts. Spongy mesophyll air spaces increase internal exchange area. Stomata allow gas exchange; guard cells regulate opening. Xylem supplies water and phloem transports sugars.

BIO-0047

RESPUESTA

FagusED · Biology
06 · Plant nutrition

Photosynthesis transfers light energy into chemical energy by producing glucose from carbon dioxide and water; oxygen is released. Chlorophyll absorbs light. Glucose may be respired, converted to starch or cellulose, combined with nitrate to make amino acids, or transported as sucrose.

BIO-0046

RESPUESTA

FagusED · Biology
07 · Human nutrition

movement of digested molecules through the intestinal wall into blood or lymph

BIO-0051

RESPUESTA

FagusED · Biology
07 · Human nutrition

breakdown of large insoluble molecules into small soluble molecules

BIO-0050

RESPUESTA

FagusED · Biology
07 · Human nutrition

taking substances such as food and drink into the body through the mouth

BIO-0049

RESPUESTA

FagusED · Biology
07 · Human nutrition

A balanced diet supplies carbohydrate, protein, lipid, vitamins, minerals, fibre and water in proportions matched to age, sex and activity. Energy imbalance can cause obesity or mass loss. Vitamin C prevents scurvy; vitamin D and calcium support bones; iron is needed for haemoglobin.

BIO-0054

RESPUESTA

FagusED · Biology
07 · Human nutrition

passing undigested food out through the anus

BIO-0053

RESPUESTA

FagusED · Biology
07 · Human nutrition

incorporation of absorbed nutrients into cells and tissues

BIO-0052

RESPUESTA

FagusED · Biology
08 · Transport in plants

loss of water vapour from plant leaves

BIO-0057

RESPUESTA

FagusED · Biology
07 · Human nutrition

nutrient absorption decreases. surface area and transport into capillaries or lacteals are reduced

BIO-0056

RESPUESTA

FagusED · Biology
07 · Human nutrition

Mechanical digestion increases surface area without changing molecules. Chemical digestion uses enzymes to hydrolyse insoluble large molecules into small soluble molecules: amylase makes maltose, proteases make amino acids and lipases make fatty acids and glycerol.

BIO-0055

RESPUESTA

FagusED · Biology
08 · Transport in plants

living tissue transporting assimilates between sources and sinks

BIO-0060

RESPUESTA

FagusED · Biology
08 · Transport in plants

tissue transporting water and mineral ions and providing support

BIO-0059

RESPUESTA

FagusED · Biology
08 · Transport in plants

movement of sucrose and amino acids in phloem

BIO-0058

RESPUESTA

FagusED · Biology
08 · Transport in plants

Define: source.

BIO-0061

PREGUNTA

FagusED · Biology
08 · Transport in plants

Explain the key idea of xylem.

BIO-0062

PREGUNTA

FagusED · Biology
08 · Transport in plants

Explain the key idea of water uptake.

BIO-0063

PREGUNTA

FagusED · Biology
08 · Transport in plants

Apply the concept: stomata close during a hot dry afternoon. What would you conclude?

BIO-0064

PREGUNTA

FagusED · Biology
09 · Transport in animals

Define: artery.

BIO-0065

PREGUNTA

FagusED · Biology
09 · Transport in animals

Define: vein.

BIO-0066

PREGUNTA

FagusED · Biology
09 · Transport in animals

Define: capillary.

BIO-0067

PREGUNTA

FagusED · Biology
09 · Transport in animals

Define: haemoglobin.

BIO-0068

PREGUNTA

FagusED · Biology
09 · Transport in animals

Define: coronary artery.

BIO-0069

PREGUNTA

FagusED · Biology
09 · Transport in animals

Explain the key idea of double circulation.

BIO-0070

PREGUNTA

FagusED · Biology
09 · Transport in animals

Explain the key idea of heart function.

BIO-0071

PREGUNTA

FagusED · Biology
09 · Transport in animals

Apply the concept: a coronary artery becomes blocked. What would you conclude?

BIO-0072

PREGUNTA

FagusED · Biology
10 · Diseases and immunity

Define: pathogen.

BIO-0073

PREGUNTA

FagusED · Biology
10 · Diseases and immunity

Define: antigen.

BIO-0074

PREGUNTA

FagusED · Biology
10 · Diseases and immunity

Define: antibody.

BIO-0075

PREGUNTA

FagusED · Biology
08 · Transport in plants

Root hair cells provide a large surface area. Water enters by osmosis from higher soil water potential to lower cell water potential. Mineral ions may enter by active transport. Water crosses the cortex and enters xylem, maintaining a continuous column.

BIO-0063

RESPUESTA

FagusED · Biology
08 · Transport in plants

Xylem vessels are dead, hollow tubes with lignified walls. They carry water and mineral ions upward and also support the plant. No end walls and a continuous lumen reduce resistance; lignin prevents collapse under tension.

BIO-0062

RESPUESTA

FagusED · Biology
08 · Transport in plants

a region releasing sucrose into phloem

BIO-0061

RESPUESTA

FagusED · Biology
09 · Transport in animals

a vessel carrying blood toward the heart

BIO-0066

RESPUESTA

FagusED · Biology
09 · Transport in animals

a blood vessel carrying blood away from the heart

BIO-0065

RESPUESTA

FagusED · Biology
08 · Transport in plants

water loss decreases but photosynthesis may fall. smaller stomatal opening reduces diffusion of water vapour and also limits carbon dioxide entry

BIO-0064

RESPUESTA

FagusED · Biology
09 · Transport in animals

a vessel supplying the heart muscle with oxygenated blood

BIO-0069

RESPUESTA

FagusED · Biology
09 · Transport in animals

a red-cell protein that combines reversibly with oxygen

BIO-0068

RESPUESTA

FagusED · Biology
09 · Transport in animals

a microscopic vessel with a one-cell-thick wall for exchange

BIO-0067

RESPUESTA

FagusED · Biology
09 · Transport in animals

cardiac muscle may die and heart output fall. reduced oxygen delivery limits aerobic respiration in cardiac muscle

BIO-0072

RESPUESTA

FagusED · Biology
09 · Transport in animals

The right side pumps deoxygenated blood to lungs; the left side pumps oxygenated blood to the body. Valves prevent backflow. The left ventricle has the thickest wall because it produces high systemic pressure. Coronary arteries supply cardiac muscle.

BIO-0071

RESPUESTA

FagusED · Biology
09 · Transport in animals

Humans have a closed double circulation: heart to lungs and back, then heart to body and back. Separation maintains low pulmonary pressure to protect lungs and high systemic pressure for rapid delivery. Fish have a single circulation.

BIO-0070

RESPUESTA

FagusED · Biology
10 · Diseases and immunity

a specific protein produced by lymphocytes that binds an antigen

BIO-0075

RESPUESTA

FagusED · Biology
10 · Diseases and immunity

a molecule recognised as foreign that stimulates an immune response

BIO-0074

RESPUESTA

FagusED · Biology
10 · Diseases and immunity

an organism that causes disease

BIO-0073

RESPUESTA

FagusED · Biology
10 · Diseases and immunity

Define: immunity.

BIO-0076

PREGUNTA

FagusED · Biology
10 · Diseases and immunity

Define: vaccination.

BIO-0077

PREGUNTA

FagusED · Biology
10 · Diseases and immunity

Explain the key idea of pathogens and transmission.

BIO-0078

PREGUNTA

FagusED · Biology
10 · Diseases and immunity

Explain the key idea of first-line defences.

BIO-0079

PREGUNTA

FagusED · Biology
10 · Diseases and immunity

Apply the concept: vaccination coverage falls in a community. What would you conclude?

BIO-0080

PREGUNTA

FagusED · Biology
11 · Gas exchange in humans

Define: ventilation.

BIO-0081

PREGUNTA

FagusED · Biology
11 · Gas exchange in humans

Define: gas exchange.

BIO-0082

PREGUNTA

FagusED · Biology
11 · Gas exchange in humans

Define: alveolus.

BIO-0083

PREGUNTA

FagusED · Biology
11 · Gas exchange in humans

Define: diaphragm.

BIO-0084

PREGUNTA

FagusED · Biology
11 · Gas exchange in humans

Define: emphysema.

BIO-0085

PREGUNTA

FagusED · Biology
11 · Gas exchange in humans

Explain the key idea of gas exchange system.

BIO-0086

PREGUNTA

FagusED · Biology
11 · Gas exchange in humans

Explain the key idea of alveolar adaptations.

BIO-0087

PREGUNTA

FagusED · Biology
11 · Gas exchange in humans

Apply the concept: alveolar walls are destroyed by smoking. What would you conclude?

BIO-0088

PREGUNTA

FagusED · Biology
12 · Respiration

Define: respiration.

BIO-0089

PREGUNTA

FagusED · Biology
12 · Respiration

Define: aerobic respiration.

BIO-0090

PREGUNTA

FagusED · Biology
10 · Diseases and immunity

A pathogen is a disease-causing organism and may be a bacterium, virus, fungus or protoctist. Transmission can be direct, by droplets or body fluids, or indirect through food, water, surfaces or vectors. Good sanitation, safe water, hygiene and vector control break transmission routes.

BIO-0078

RESPUESTA

FagusED · Biology
10 · Diseases and immunity

introduction of harmless antigenic material to produce immunity

BIO-0077

RESPUESTA

FagusED · Biology
10 · Diseases and immunity

the ability to resist a particular infection through an effective response

BIO-0076

RESPUESTA

FagusED · Biology
11 · Gas exchange in humans

movement of air into and out of the lungs

BIO-0081

RESPUESTA

FagusED · Biology
10 · Diseases and immunity

outbreak risk rises. fewer immune people allow longer transmission chains and reduce indirect protection

BIO-0080

RESPUESTA

FagusED · Biology
10 · Diseases and immunity

Skin forms a physical barrier and clotting seals wounds. Nasal hairs and mucus trap pathogens; cilia move mucus toward the throat. Stomach acid kills many swallowed microorganisms. These non-specific defences act before an immune response.

BIO-0079

RESPUESTA

FagusED · Biology
11 · Gas exchange in humans

a sheet of muscle below the lungs that changes thoracic volume

BIO-0084

RESPUESTA

FagusED · Biology
11 · Gas exchange in humans

a microscopic air sac where gas exchange occurs

BIO-0083

RESPUESTA

FagusED · Biology
11 · Gas exchange in humans

diffusion of oxygen and carbon dioxide between alveoli and blood

BIO-0082

RESPUESTA

FagusED · Biology
11 · Gas exchange in humans

Millions of alveoli provide large area. Their walls and capillary walls are one cell thick and moist. Dense blood flow and ventilation maintain steep concentration gradients. Red cells rapidly bind oxygen, helping preserve the gradient.

BIO-0087

RESPUESTA

FagusED · Biology
11 · Gas exchange in humans

Air passes through trachea, bronchi and bronchioles to alveoli. Cartilage keeps larger airways open; mucus traps particles and cilia move it away. Gas exchange is diffusion of oxygen into blood and carbon dioxide out.

BIO-0086

RESPUESTA

FagusED · Biology
11 · Gas exchange in humans

damage to alveolar walls that reduces surface area and elasticity

BIO-0085

RESPUESTA

FagusED · Biology
12 · Respiration

energy-releasing breakdown of nutrients using oxygen

BIO-0090

RESPUESTA

FagusED · Biology
12 · Respiration

chemical reactions in cells that break down nutrient molecules and release energy

BIO-0089

RESPUESTA

FagusED · Biology
11 · Gas exchange in humans

diffusion capacity and oxygen uptake decrease. surface area falls and ventilation becomes less effective due to loss of elasticity

BIO-0088

RESPUESTA

FagusED · Biology
12 · Respiration

Define: anaerobic respiration.

BIO-0091

PREGUNTA

FagusED · Biology
12 · Respiration

Define: oxygen debt.

BIO-0092

PREGUNTA

FagusED · Biology
12 · Respiration

Define: fermentation.

BIO-0093

PREGUNTA

FagusED · Biology
12 · Respiration

Explain the key idea of aerobic respiration.

BIO-0094

PREGUNTA

FagusED · Biology
12 · Respiration

Explain the key idea of anaerobic respiration.

BIO-0095

PREGUNTA

FagusED · Biology
12 · Respiration

Apply the concept: a sprinter continues breathing rapidly after finishing. What would you conclude?

BIO-0096

PREGUNTA

FagusED · Biology
13 · Excretion in humans

Define: excretion.

BIO-0097

PREGUNTA

FagusED · Biology
13 · Excretion in humans

Define: urea.

BIO-0098

PREGUNTA

FagusED · Biology
13 · Excretion in humans

Define: ultrafiltration.

BIO-0099

PREGUNTA

FagusED · Biology
13 · Excretion in humans

Define: selective reabsorption.

BIO-0100

PREGUNTA

FagusED · Biology
13 · Excretion in humans

Define: dialysis.

BIO-0101

PREGUNTA

FagusED · Biology
13 · Excretion in humans

Explain the key idea of excretion.

BIO-0102

PREGUNTA

FagusED · Biology
13 · Excretion in humans

Explain the key idea of kidney and nephron.

BIO-0103

PREGUNTA

FagusED · Biology
13 · Excretion in humans

Apply the concept: a patient produces urine containing glucose. What would you conclude?

BIO-0104

PREGUNTA

FagusED · Biology
14 · Coordination and response

Define: stimulus.

BIO-0105

PREGUNTA

FagusED · Biology
14 · Coordination and response

Define: receptor.

BIO-0106

PREGUNTA

FagusED · Biology
14 · Coordination and response

Define: effector.

BIO-0107

PREGUNTA

FagusED · Biology
14 · Coordination and response

Define: homeostasis.

BIO-0108

PREGUNTA

FagusED · Biology
12 · Respiration

anaerobic respiration by microorganisms such as yeast

BIO-0093

RESPUESTA

FagusED · Biology
12 · Respiration

extra oxygen required after exercise to deal with lactic acid

BIO-0092

RESPUESTA

FagusED · Biology
12 · Respiration

energy release without oxygen

BIO-0091

RESPUESTA

FagusED · Biology
12 · Respiration

extra oxygen is required. oxygen supports removal or oxidation of lactic acid and restoration of normal conditions

BIO-0096

RESPUESTA

FagusED · Biology
12 · Respiration

Anaerobic respiration releases less energy per glucose because breakdown is incomplete. In human muscles, glucose forms lactic acid. In yeast, glucose forms ethanol and carbon dioxide; this fermentation is used in bread and alcoholic-drink production.

BIO-0095

RESPUESTA

FagusED · Biology
12 · Respiration

Aerobic respiration is the chemical reactions in cells that use oxygen to break down nutrient molecules and release energy. Glucose and oxygen form carbon dioxide and water. Energy supports muscle contraction, active transport, protein synthesis, cell division and temperature maintenance.

BIO-0094

RESPUESTA

FagusED · Biology
13 · Excretion in humans

pressure filtration of small molecules from glomerular blood

BIO-0099

RESPUESTA

FagusED · Biology
13 · Excretion in humans

a nitrogenous waste made in the liver from excess amino acids

BIO-0098

RESPUESTA

FagusED · Biology
13 · Excretion in humans

removal of metabolic waste, toxic material and excess substances

BIO-0097

RESPUESTA

FagusED · Biology
13 · Excretion in humans

Excretion removes toxic materials, waste products of metabolism and substances in excess. Carbon dioxide is excreted by lungs; urea is formed in the liver from excess amino acids and excreted by kidneys; excess water and ions leave through kidneys and skin. Egestion is removal of undigested food and is not excretion.

BIO-0102

RESPUESTA

FagusED · Biology
13 · Excretion in humans

artificial removal of wastes and excess ions or water across a membrane

BIO-0101

RESPUESTA

FagusED · Biology
13 · Excretion in humans

return of useful substances from nephron filtrate to blood

BIO-0100

RESPUESTA

FagusED · Biology
14 · Coordination and response

a change in the internal or external environment detected by a receptor

BIO-0105

RESPUESTA

FagusED · Biology
13 · Excretion in humans

selective reabsorption capacity may be exceeded or impaired. normally all filtered glucose is reabsorbed into blood

BIO-0104

RESPUESTA

FagusED · Biology
13 · Excretion in humans

Renal arteries deliver blood; renal veins remove filtered blood; ureters carry urine to the bladder; the urethra releases urine. Each nephron contains a glomerulus, Bowman's capsule, tubule and collecting duct.

BIO-0103

RESPUESTA

FagusED · Biology
14 · Coordination and response

maintenance of a stable internal environment

BIO-0108

RESPUESTA

FagusED · Biology
14 · Coordination and response

a muscle or gland that carries out a response

BIO-0107

RESPUESTA

FagusED · Biology
14 · Coordination and response

a cell or sense organ detecting a stimulus

BIO-0106

RESPUESTA

FagusED · Biology
14 · Coordination and response

Define: reflex.

BIO-0109

PREGUNTA

FagusED · Biology
14 · Coordination and response

Explain the key idea of nervous system.

BIO-0110

PREGUNTA

FagusED · Biology
14 · Coordination and response

Explain the key idea of eye.

BIO-0111

PREGUNTA

FagusED · Biology
14 · Coordination and response

Apply the concept: blood glucose rises after a meal. What would you conclude?

BIO-0112

PREGUNTA

FagusED · Biology
15 · Drugs

Define: drug.

BIO-0113

PREGUNTA

FagusED · Biology
15 · Drugs

Define: antibiotic.

BIO-0114

PREGUNTA

FagusED · Biology
15 · Drugs

Define: tolerance.

BIO-0115

PREGUNTA

FagusED · Biology
15 · Drugs

Define: dependence.

BIO-0116

PREGUNTA

FagusED · Biology
15 · Drugs

Define: placebo.

BIO-0117

PREGUNTA

FagusED · Biology
15 · Drugs

Explain the key idea of medicinal drugs.

BIO-0118

PREGUNTA

FagusED · Biology
15 · Drugs

Explain the key idea of antibiotics.

BIO-0119

PREGUNTA

FagusED · Biology
15 · Drugs

Apply the concept: a person stops an addictive depressant suddenly. What would you conclude?

BIO-0120

PREGUNTA

FagusED · Biology
16 · Reproduction

Define: pollination.

BIO-0121

PREGUNTA

FagusED · Biology
16 · Reproduction

Define: fertilisation.

BIO-0122

PREGUNTA

FagusED · Biology
16 · Reproduction

Define: implantation.

BIO-0123

PREGUNTA

FagusED · Biology
14 · Coordination and response

The cornea and lens refract light onto the retina; the iris controls pupil size; rods detect low light and cones detect colour. For near objects, ciliary muscles contract, suspensory ligaments slacken and the lens becomes thicker and more refractive.

BIO-0111

RESPUESTA

FagusED · Biology
14 · Coordination and response

Receptors detect stimuli; impulses travel along sensory neurones to the central nervous system, across synapses using neurotransmitters, then along motor neurones to effectors. Reflexes are rapid automatic responses that protect the body.

BIO-0110

RESPUESTA

FagusED · Biology
14 · Coordination and response

a rapid automatic response to a stimulus

BIO-0109

RESPUESTA

FagusED · Biology
15 · Drugs

a medicine that kills bacteria or inhibits bacterial growth

BIO-0114

RESPUESTA

FagusED · Biology
15 · Drugs

a substance taken into the body that modifies chemical reactions

BIO-0113

RESPUESTA

FagusED · Biology
14 · Coordination and response

insulin secretion increases and glucose falls toward normal. cells take up glucose and liver or muscle convert it to glycogen by negative feedback

BIO-0112

RESPUESTA

FagusED · Biology
15 · Drugs

an inactive treatment used as a comparison

BIO-0117

RESPUESTA

FagusED · Biology
15 · Drugs

a state in which stopping a drug causes withdrawal

BIO-0116

RESPUESTA

FagusED · Biology
15 · Drugs

needing a larger dose for the same effect

BIO-0115

RESPUESTA

FagusED · Biology
15 · Drugs

withdrawal symptoms may occur. dependence has altered normal nervous-system function and the drug is no longer present

BIO-0120

RESPUESTA

FagusED · Biology
15 · Drugs

Antibiotics kill bacteria or inhibit their growth but do not treat viral infections. Random mutations create variation in bacterial populations; antibiotic exposure kills susceptible cells while resistant bacteria survive, reproduce and spread resistance. Misuse increases selection pressure.

BIO-0119

RESPUESTA

FagusED · Biology
15 · Drugs

A drug is a substance taken into the body that modifies chemical reactions. Medicinal drugs prevent or treat disease or relieve symptoms and should be tested for effectiveness, safe dose and side effects. Placebo-controlled, double-blind trials reduce bias.

BIO-0118

RESPUESTA

FagusED · Biology
16 · Reproduction

attachment of an early embryo to the uterine lining

BIO-0123

RESPUESTA

FagusED · Biology
16 · Reproduction

fusion of nuclei of male and female gametes

BIO-0122

RESPUESTA

FagusED · Biology
16 · Reproduction

transfer of pollen from anther to stigma

BIO-0121

RESPUESTA

FagusED · Biology
16 · Reproduction

Define: gamete.

BIO-0124

PREGUNTA

FagusED · Biology
16 · Reproduction

Define: asexual reproduction.

BIO-0125

PREGUNTA

FagusED · Biology
16 · Reproduction

Explain the key idea of asexual and sexual reproduction.

BIO-0126

PREGUNTA

FagusED · Biology
16 · Reproduction

Explain the key idea of flowering plants.

BIO-0127

PREGUNTA

FagusED · Biology
16 · Reproduction

Apply the concept: an insect-pollinated flower population loses pollinators. What would you conclude?

BIO-0128

PREGUNTA

FagusED · Biology
17 · Inheritance

Define: gene.

BIO-0129

PREGUNTA

FagusED · Biology
17 · Inheritance

Define: allele.

BIO-0130

PREGUNTA

FagusED · Biology
17 · Inheritance

Define: genotype.

BIO-0131

PREGUNTA

FagusED · Biology
17 · Inheritance

Define: phenotype.

BIO-0132

PREGUNTA

FagusED · Biology
17 · Inheritance

Define: heterozygous.

BIO-0133

PREGUNTA

FagusED · Biology
17 · Inheritance

Explain the key idea of genetic information.

BIO-0134

PREGUNTA

FagusED · Biology
17 · Inheritance

Explain the key idea of mitosis.

BIO-0135

PREGUNTA

FagusED · Biology
17 · Inheritance

Apply the concept: two heterozygous parents carry a recessive condition. What would you conclude?

BIO-0136

PREGUNTA

FagusED · Biology
18 · Variation and selection

Define: variation.

BIO-0137

PREGUNTA

FagusED · Biology
18 · Variation and selection

Define: mutation.

BIO-0138

PREGUNTA

FagusED · Biology
16 · Reproduction

Asexual reproduction uses one parent, no gamete fusion and produces genetically identical offspring except for mutation. It is rapid and effective in stable conditions. Sexual reproduction fuses haploid gametes, produces variation and may improve adaptation when conditions change, but requires more time and energy.

BIO-0126

RESPUESTA

FagusED · Biology
16 · Reproduction

reproduction from one parent without gamete fusion

BIO-0125

RESPUESTA

FagusED · Biology
16 · Reproduction

a haploid reproductive cell that fuses during fertilisation

BIO-0124

RESPUESTA

FagusED · Biology
17 · Inheritance

a length of DNA that codes for a protein

BIO-0129

RESPUESTA

FagusED · Biology
16 · Reproduction

seed production decreases. fewer pollen grains reach compatible stigmas, reducing fertilisation

BIO-0128

RESPUESTA

FagusED · Biology
16 · Reproduction

Anthers make pollen and ovaries contain ovules. Pollination transfers pollen from anther to stigma; fertilisation fuses male nucleus with egg nucleus in an ovule. A pollen tube grows through the style. After fertilisation, ovule becomes seed and ovary becomes fruit.

BIO-0127

RESPUESTA

FagusED · Biology
17 · Inheritance

observable features resulting from genotype and environment

BIO-0132

RESPUESTA

FagusED · Biology
17 · Inheritance

the alleles an organism possesses

BIO-0131

RESPUESTA

FagusED · Biology
17 · Inheritance

an alternative form of a gene

BIO-0130

RESPUESTA

FagusED · Biology
17 · Inheritance

Before mitosis DNA replicates. Nuclear division produces genetically identical nuclei with the same chromosome number, followed by cell division. Mitosis enables growth, replacement, repair and asexual reproduction. Uncontrolled division can form tumours.

BIO-0135

RESPUESTA

FagusED · Biology
17 · Inheritance

A chromosome is a DNA molecule carrying genes. A gene is a length of DNA coding for a protein; an allele is an alternative form of a gene. Base sequence determines amino-acid sequence and protein shape. Most body cells are diploid with homologous chromosome pairs.

BIO-0134

RESPUESTA

FagusED · Biology
17 · Inheritance

having two different alleles of a gene

BIO-0133

RESPUESTA

FagusED · Biology
18 · Variation and selection

a random change in genetic material

BIO-0138

RESPUESTA

FagusED · Biology
18 · Variation and selection

differences between individuals of the same species

BIO-0137

RESPUESTA

FagusED · Biology
17 · Inheritance

each child has a 25% probability of being affected. the cross gives AA, Aa, Aa and aa in a 1:2:1 genotype ratio

BIO-0136

RESPUESTA

FagusED · Biology
18 · Variation and selection

Define: adaptation.

BIO-0139

PREGUNTA

FagusED · Biology
18 · Variation and selection

Define: natural selection.

BIO-0140

PREGUNTA

FagusED · Biology
18 · Variation and selection

Define: selective breeding.

BIO-0141

PREGUNTA

FagusED · Biology
18 · Variation and selection

Explain the key idea of variation.

BIO-0142

PREGUNTA

FagusED · Biology
18 · Variation and selection

Explain the key idea of mutation.

BIO-0143

PREGUNTA

FagusED · Biology
18 · Variation and selection

Apply the concept: a drought persists for many generations. What would you conclude?

BIO-0144

PREGUNTA

FagusED · Biology
19 · Organisms and their environment

Define: population.

BIO-0145

PREGUNTA

FagusED · Biology
19 · Organisms and their environment

Define: community.

BIO-0146

PREGUNTA

FagusED · Biology
19 · Organisms and their environment

Define: ecosystem.

BIO-0147

PREGUNTA

FagusED · Biology
19 · Organisms and their environment

Define: trophic level.

BIO-0148

PREGUNTA

FagusED · Biology
19 · Organisms and their environment

Define: carrying capacity.

BIO-0149

PREGUNTA

FagusED · Biology
19 · Organisms and their environment

Explain the key idea of energy flow.

BIO-0150

PREGUNTA

FagusED · Biology
19 · Organisms and their environment

Explain the key idea of food chains and pyramids.

BIO-0151

PREGUNTA

FagusED · Biology
19 · Organisms and their environment

Apply the concept: a top predator is removed from a food web. What would you conclude?

BIO-0152

PREGUNTA

FagusED · Biology
20 · Human influences on ecosystems

Define: biodiversity.

BIO-0153

PREGUNTA

FagusED · Biology
18 · Variation and selection

human selection of parents with desired features

BIO-0141

RESPUESTA

FagusED · Biology
18 · Variation and selection

differential survival and reproduction caused by environmental pressures

BIO-0140

RESPUESTA

FagusED · Biology
18 · Variation and selection

an inherited feature increasing fitness in an environment

BIO-0139

RESPUESTA

FagusED · Biology
18 · Variation and selection

alleles for water-conserving features may increase. individuals with advantageous inherited variation survive and reproduce more

BIO-0144

RESPUESTA

FagusED · Biology
18 · Variation and selection

Mutation is a random change in genetic material. Gene mutations alter DNA base sequence; chromosome mutations change structure or number. Mutations can be harmful, neutral or beneficial depending on environment. Ionising radiation and some chemicals increase mutation rate.

BIO-0143

RESPUESTA

FagusED · Biology
18 · Variation and selection

Continuous variation shows a range and is often controlled by many genes plus environment, such as height. Discontinuous variation has distinct categories and is often controlled by one or few genes, such as ABO blood group. Genetic variation arises from mutation, meiosis and fertilisation.

BIO-0142

RESPUESTA

FagusED · Biology
19 · Organisms and their environment

a community and its interactions with the physical environment

BIO-0147

RESPUESTA

FagusED · Biology
19 · Organisms and their environment

all populations of different species in an area

BIO-0146

RESPUESTA

FagusED · Biology
19 · Organisms and their environment

all organisms of one species in an area at a given time

BIO-0145

RESPUESTA

FagusED · Biology
19 · Organisms and their environment

The Sun is the main energy source. Producers transfer light energy into chemical energy; consumers obtain it by feeding. Energy is lost between trophic levels in respiration, movement, heat, waste and uneaten material, so food chains are short and biomass decreases upward.

BIO-0150

RESPUESTA

FagusED · Biology
19 · Organisms and their environment

the maximum population an environment can sustain

BIO-0149

RESPUESTA

FagusED · Biology
19 · Organisms and their environment

the feeding position of an organism in a food chain

BIO-0148

RESPUESTA

FagusED · Biology
20 · Human influences on ecosystems

the variety of genes, species and ecosystems in an area

BIO-0153

RESPUESTA

FagusED · Biology
19 · Organisms and their environment

its prey may increase and producers may decrease. reduced predation changes population balance and can create a trophic cascade

BIO-0152

RESPUESTA

FagusED · Biology
19 · Organisms and their environment

Arrows in a food chain show direction of energy transfer. Food webs reveal interdependence. Pyramids of biomass usually better represent available living material than numbers. Efficiency is energy transferred to the next level divided by energy available at the previous level.

BIO-0151

RESPUESTA

FagusED · Biology
20 · Human influences on ecosystems

Define: bioaccumulation.

BIO-0154

PREGUNTA

FagusED · Biology
20 · Human influences on ecosystems

Define: biomagnification.

BIO-0155

PREGUNTA

FagusED · Biology
20 · Human influences on ecosystems

Define: conservation.

BIO-0156

PREGUNTA

FagusED · Biology
20 · Human influences on ecosystems

Define: sustainable resource.

BIO-0157

PREGUNTA

FagusED · Biology
20 · Human influences on ecosystems

Explain the key idea of food production.

BIO-0158

PREGUNTA

FagusED · Biology
20 · Human influences on ecosystems

Explain the key idea of habitat destruction.

BIO-0159

PREGUNTA

FagusED · Biology
20 · Human influences on ecosystems

Apply the concept: a forest is divided by roads into small fragments. What would you conclude?

BIO-0160

PREGUNTA

FagusED · Biology
21 · Biotechnology and genetic modification

Define: biotechnology.

BIO-0161

PREGUNTA

FagusED · Biology
21 · Biotechnology and genetic modification

Define: fermenter.

BIO-0162

PREGUNTA

FagusED · Biology
21 · Biotechnology and genetic modification

Define: plasmid.

BIO-0163

PREGUNTA

FagusED · Biology
21 · Biotechnology and genetic modification

Define: restriction enzyme.

BIO-0164

PREGUNTA

FagusED · Biology
21 · Biotechnology and genetic modification

Define: ligase.

BIO-0165

PREGUNTA

FagusED · Biology
21 · Biotechnology and genetic modification

Explain the key idea of microorganisms in biotechnology.

BIO-0166

PREGUNTA

FagusED · Biology
21 · Biotechnology and genetic modification

Explain the key idea of industrial enzymes.

BIO-0167

PREGUNTA

FagusED · Biology
21 · Biotechnology and genetic modification

Apply the concept: a pest-resistant GM crop is widely planted. What would you conclude?

BIO-0168

PREGUNTA

FagusED · Biology**20 · Human influences on ecosystems**

management and protection of biodiversity and resources

BIO-0156

RESPUESTA

FagusED · Biology**20 · Human influences on ecosystems**

increase in pollutant concentration at successive trophic levels

BIO-0155

RESPUESTA

FagusED · Biology**20 · Human influences on ecosystems**

build-up of a substance within an organism over time

BIO-0154

RESPUESTA

FagusED · Biology**20 · Human influences on ecosystems**

Deforestation removes habitats, reduces biodiversity, exposes soil to erosion and decreases carbon storage. Land is cleared for timber, agriculture, mining and settlement. Fragmentation isolates populations, reduces gene flow and increases extinction risk.

BIO-0159

RESPUESTA

FagusED · Biology**20 · Human influences on ecosystems**

Agricultural yield can increase through fertilisers, irrigation, pesticides, selective breeding, machinery and controlled environments. Intensive systems improve output but may reduce biodiversity, pollute water, select resistant pests and affect animal welfare. Sustainable approaches balance yield with long-term soil and ecosystem health.

BIO-0158

RESPUESTA

FagusED · Biology**20 · Human influences on ecosystems**

a resource used at a rate that allows replacement and long-term availability

BIO-0157

RESPUESTA

FagusED · Biology**21 · Biotechnology and genetic modification**

a controlled vessel for large-scale growth of microorganisms

BIO-0162

RESPUESTA

FagusED · Biology**21 · Biotechnology and genetic modification**

use of organisms, cells or enzymes to make useful products

BIO-0161

RESPUESTA

FagusED · Biology**20 · Human influences on ecosystems**

local extinction risk increases. populations become smaller and isolated, reducing gene flow and access to resources

BIO-0160

RESPUESTA

FagusED · Biology**21 · Biotechnology and genetic modification**

an enzyme that joins DNA fragments

BIO-0165

RESPUESTA

FagusED · Biology**21 · Biotechnology and genetic modification**

an enzyme that cuts DNA at specific sequences

BIO-0164

RESPUESTA

FagusED · Biology**21 · Biotechnology and genetic modification**

a small circular DNA molecule used as a vector

BIO-0163

RESPUESTA

FagusED · Biology**21 · Biotechnology and genetic modification**

insecticide use may fall but selection for resistant pests may increase. the inserted gene protects plants, while strong selection favours rare resistant insects

BIO-0168

RESPUESTA

FagusED · Biology**21 · Biotechnology and genetic modification**

Pectinase increases fruit-juice yield and clarity; lactase makes lactose-free milk; biological washing powders contain proteases and lipases. Immobilised enzymes can be reused, are easy to separate from product and often tolerate wider conditions.

BIO-0167

RESPUESTA

FagusED · Biology**21 · Biotechnology and genetic modification**

Microorganisms grow rapidly, use varied substrates and can produce enzymes, antibiotics and foods. Fermenters maintain temperature, pH, oxygen and nutrients while stirring distributes cells and heat. Sterile conditions prevent contamination and competition. Products are harvested and purified.

BIO-0166

RESPUESTA